

Exercise 3: Sorting Customer Orders

Aim

To implement Bubble Sort and Quick Sort algorithms in Java and compare their performance for sorting customer orders based on total price.

Problem Understanding

Sorting customer orders helps an e-commerce platform identify and process high-value orders efficiently. Different sorting algorithms have different performance characteristics, making algorithm selection important.

Sorting Algorithms

Bubble Sort

Bubble Sort repeatedly compares adjacent elements and swaps them if they are in the wrong order. It is simple to understand but becomes slow for large datasets.

Quick Sort

Quick Sort selects a pivot element and partitions the array into smaller subarrays. It recursively sorts these subarrays, making it much faster for larger collections of data.

Classes Used

Order

Stores the following details:

- Order ID
- Customer Name
- Total Price

Sorting Orders

Implements:

- Bubble Sort
- Quick Sort
- Display Function

Time Complexity

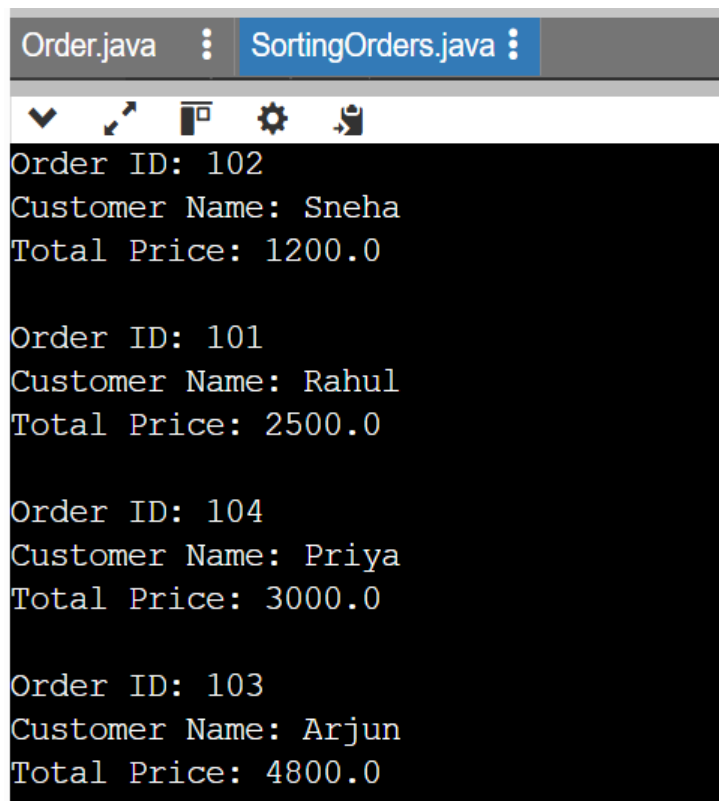
Algorithm	Best	Average	Worst
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$

Analysis

Bubble Sort performs many unnecessary comparisons and swaps, making it inefficient for large datasets.

Quick Sort divides the data into smaller parts and sorts them recursively, resulting in much better average performance. Therefore, Quick Sort is generally preferred for real-world applications such as e-commerce platforms.

Output



```

Order.java  :  SortingOrders.java  :
Order ID: 102
Customer Name: Sneha
Total Price: 1200.0

Order ID: 101
Customer Name: Rahul
Total Price: 2500.0

Order ID: 104
Customer Name: Priya
Total Price: 3000.0

Order ID: 103
Customer Name: Arjun
Total Price: 4800.0

```

```
Quick Sort Result  
Order ID: 102  
Customer Name: Sneha  
Total Price: 1200.0
```

```
Order ID: 101  
Customer Name: Rahul  
Total Price: 2500.0
```

```
Order ID: 104  
Customer Name: Priya  
Total Price: 3000.0
```

```
Order ID: 103  
Customer Name: Arjun  
Total Price: 4800.0
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```

Conclusion

This exercise demonstrates the implementation of Bubble Sort and Quick Sort for sorting customer orders. The comparison shows that Quick Sort is generally more efficient and suitable for handling large amounts of data.